

Polynomial Regression Assignment Report

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1. Objective and approach in brief

The objective is to predict the continuous target variable y for two personalised datasets using polynomial regression. `var1` contains six input variables and permits polynomial degree up to 10, while `var2` contains three input variables and permits degree up to 20. The main challenge is to choose a model that captures the nonlinear relationship without unnecessary overfitting.

The final implementation does not assume that Ridge is always better. For each dataset, it automatically compares Ridge and Lasso regularisation while also searching the polynomial degree and the regularisation strength α . The combination with the lowest 5-fold cross-validation mean squared error (MSE) is selected.

Final selected models

Problem	Selected method	Degree	Alpha	CV MSE	Polynomial terms
var1	Lasso	5	0.01	0.3251	461
var2	Ridge	10	1	0.2509	285

2. Data overview

Property	var1	var2
Training / test rows	1000 / 1000	1000 / 1000
Number of inputs	6 ($x_1 \dots x_6$)	3 (x_1, x_2, x_3)
Input range	$[-1, 1]$	$[-1, 1]$
Maximum degree allowed	10	20

Both datasets contain 1000 training observations and 1000 test observations. The input variables are already bounded in the interval $[-1, 1]$. The target y is available only in the training data; the test targets are hidden and are therefore not used in model selection.

3. Method

3.1 Polynomial feature expansion

For every candidate degree d , `PolynomialFeatures` generates polynomial powers and interaction terms whose total power is at most d . `include_bias=False` is used, so no separate bias column is introduced by the polynomial expansion.

3.2 Standardisation

The generated polynomial features are standardised with `StandardScaler` before regularised regression. All transformations are contained in the same Pipeline as the regression model, so the preprocessing is fitted separately within each training fold during cross-validation.

3.3 Ridge and Lasso

Two regularisation methods are considered. Ridge applies an L2 penalty and tends to shrink correlated polynomial coefficients together. Lasso applies an L1 penalty and can drive some coefficients exactly to zero, which can be useful when many generated terms are unnecessary. The model type is therefore treated as a hyperparameter rather than being fixed in advance.

4. Automatic model selection

4.1 Search space

The code searches every allowed polynomial degree and seven logarithmically spaced alpha values. The alpha candidates are generated by `10.0 ** np.arange(-3, 4)`, giving 0.001, 0.01, 0.1, 1, 10, 100 and 1000. Both Ridge and Lasso are tested for every degree-alpha combination.

Parameter	var1	var2
Polynomial degree searched	1 to 10	1 to 20
Regression types	Ridge, Lasso	Ridge, Lasso
Alpha candidates	0.001 to 1000	0.001 to 1000
Validation	5-fold KFold, shuffled	5-fold KFold, shuffled
Selection metric	Mean CV MSE	Mean CV MSE

4.2 Five-fold cross-validation

KFold is configured with `n_splits=5`, `shuffle=True` and `random_state=42`. For each candidate model, four folds are used for training and the remaining fold is used for validation. This is repeated so that every fold acts as the validation fold once. The mean validation MSE is then used by GridSearchCV to rank the candidate models.

4.3 Number of fits

For var1, the search contains 10 degrees x 7 alpha values x 2 model types x 5 folds = 700 validation fits. For var2, it contains 20 degrees x 7 alpha values x 2 model types x 5 folds = 1400 validation fits. This exhaustive grid is performed only on the training data.

4.4 Final fitting and prediction

GridSearchCV uses `refit=True`. After selecting the best degree, model type and alpha, the winning pipeline is refitted on all 1000 training rows for that problem. The final model is then used to predict y for every row of the corresponding test file. The test targets are never used during this procedure.

5. Results for var1 (six inputs)

For var1, Lasso clearly improves the validation error compared with the best Ridge model. The best Ridge configuration gives a CV MSE of about 0.5495 at degree 5 and alpha 10, whereas the best Lasso configuration gives a substantially lower CV MSE of about 0.3251 at degree 5 and alpha 0.01. Since lower MSE is better, Lasso is selected for var1.

Model	Degree	Alpha	Best CV MSE
Ridge	5	10	0.5495
Lasso	5	0.01	0.3251

The selected degree is 5 rather than the maximum allowed degree of 10. The comparison indicates that increasing polynomial complexity beyond the useful region does not improve the validation score enough to justify the extra flexibility.

6. Results for var2 (three inputs)

The var2 dataset behaves differently. The best Ridge model reaches a CV MSE of about 0.2509 at degree 10 and alpha 1. The best Lasso configurations in the same region remain slightly worse, around 0.257, so Ridge is selected for var2. Therefore, the optimal regularisation method can differ across the personalised datasets.

Model	Selected / best degree region	Alpha	Best CV MSE
Ridge	10	1	0.2509
Lasso	about 11	0.001	about 0.2572

The selected degree of 10 is well below the allowed maximum of 20. The validation behaviour becomes relatively flat in the neighbourhood around degrees 8-12, but degree 10 gives the lowest CV MSE found by the submitted search procedure and is therefore retained.

7. Why the selected regularisation methods differ

7.1 var1 - Lasso

At the selected degree, the polynomial expansion produces 461 terms. Lasso can shrink some of these coefficients all the way to zero, which is useful when only a subset of polynomial and interaction terms contribute strongly to prediction. The much lower CV MSE observed for Lasso indicates that this sparse regularisation is better suited to var1 among the models tested.

7.2 var2 - Ridge

For var2 at degree 10, the polynomial expansion contains 285 terms. The best Ridge model has slightly lower CV MSE than the best Lasso model. Ridge therefore provides the better balance between fitting the nonlinear relationship and controlling coefficient magnitude for this dataset.

8. Underfitting and overfitting

Low polynomial degrees can underfit the nonlinear relationship because they cannot represent enough curvature or interaction effects. As degree increases, the model becomes more expressive but also more prone to fitting noise. Cross-validation provides a practical way to locate the region where unseen-data error is minimised. The selected degrees, 5 for var1 and 10 for var2, are therefore based on validation performance rather than simply using the maximum permitted degree.

9. Evaluation

Mean squared error is the primary selection metric in the code. MSE is suitable for this task because the assignment evaluates prediction accuracy and penalises larger errors more strongly. Training MSE and training R2 are also calculated after fitting, but they are only diagnostics because they are measured on data that the final model has already seen. The hidden test MSE and R2 cannot be calculated locally.

Problem	Method	Degree	Alpha	CV MSE
var1	Lasso	5	0.01	0.3251
var2	Ridge	10	1	0.2509

10. Final models and prediction generation

After model selection, the winning pipeline for each problem is refitted using all available training observations. The resulting models are then applied to the separate test datasets. Each test file contains 1000 rows, so 1000 predictions are generated for each problem.

Problem	Method	Degree	Alpha	Polynomial terms	Output file
var1	Lasso	5	0.01	461	BT2024113_predicted_var1.csv

var2	Ridge	10	1	285	BT2024113_pred_var2.csv
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The prediction files contain a single column named y, matching the required submission format. Because the true test targets are hidden, these predictions are the final quantities submitted for evaluation.

11. Additional observations and limitations

- The code selects regression type, polynomial degree and alpha jointly; no final degree or alpha is hardcoded as the answer.
- The search uses a finite alpha grid. A finer alpha grid could potentially produce a small improvement, but the submitted model uses the specified generated candidate set.
- Polynomial feature count grows quickly with degree. Higher degrees therefore increase both computation and the risk of overfitting.
- The test targets are hidden, so cross-validation on the training set is the only local estimate of generalisation performance available before submission.
- Only polynomial regression with Ridge or Lasso regularisation is considered; the test set is never used to select a model.

12. Reproducibility

All steps are implemented in a single Python script. The same script loads the datasets, builds the PolynomialFeatures + StandardScaler + regularised-regression pipeline, performs GridSearchCV over both Ridge and Lasso, prints the selected method/degree/alpha, refits the winning model and writes the two prediction files. The shuffled KFold split uses a fixed random state of 42 for reproducibility.

13. Conclusion

The experiments show that the best regularisation method is dataset-dependent. For var1, Lasso substantially outperforms Ridge in cross-validation, giving a CV MSE of about 0.3251 at degree 5 and alpha 0.01. For var2, Ridge gives the best validation result, with degree 10 and alpha 1 producing a CV MSE of about 0.2509. The final implementation therefore uses Lasso for var1 and Ridge for var2, with both choices obtained automatically from the same model-selection procedure.

14. Deliverables

Single Python script for automatic Ridge/Lasso, degree and alpha selection.

BT2024113_pred_var1.csv - test predictions for var1.

BT2024113_pred_var2.csv - test predictions for var2.

Github link - [Hasini-Samudrala/Polynomial-Regression](https://github.com/Hasini-Samudrala/Polynomial-Regression)